

Real Salt-Phase (SNO-II) $\theta_{12}/\Delta m_{21}^2$ Inference with **tpeanuts**

Fast Simulation of Neutrino Oscillations in Matter

MSc Thesis (Trabajo Fin de Máster) — Inference Companion Document

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This document accompanies `notebooks/inference/inference6_sno_ii.ipynb` and the `tpeanuts.detector.sno_ii/tpeanuts.inference` code packages: it documents, end to end, the real Sudbury Neutrino Observatory salt-phase (Phase II) 391-day data set, the official published observable extraction it is fitted against, the equivalent-flux forward model and correlated-covariance likelihood implemented in **tpeanuts**, the inference machinery (a new correlated-Gaussian likelihood, added to this project's shared fitting code specifically to support this document), and the resulting real $\theta_{12}/\Delta m_{21}^2/\Phi_{8B}$ measurement, compared directly against SNO's own published best fit and contours. Blue boxes (**Theoretical Foundation**) contain the physics; green boxes (**Implementation**) link that physics to actual **tpeanuts** code; orange boxes collect approximations and limitations; teal boxes (**Result**) report a concrete numerical outcome of the notebook. Text highlighted in **yellow** marks key results/equations.

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Introduction to the SNO Salt Phase

1.1 Purpose and how to read this document

This document is the second inference-side companion to the *Theoretical Foundations of tpeanuts* document: where that document develops the generic neutrino-propagation physics implemented in `core/` and `medium/`, and *Inference at Daya Bay* documents a reactor antineutrino application, this one documents a real solar-neutrino application built on a different statistical foundation than either – fitting the solar mixing angle θ_{12} , the solar mass-squared splitting Δm_{21}^2 , and the total active ^8B solar-neutrino flux Φ_{8B} to the real public data of the Sudbury Neutrino Observatory’s salt phase (Phase II). Every number quoted here that is not explicitly attributed to a fitted `tpeanuts` result is a real, published, externally verifiable quantity, traceable through the References chapter.

The five chapters are organised as follows: this chapter introduces the experiment itself – the salt-phase detector modification, its three detection reactions, and the data-taking timeline – with no reference yet to `tpeanuts`; [Chapter 2](#) catalogues the real official published tables this document’s data was transcribed from, table by table, including an honest account of which entries could and could not be read with full confidence; [Chapter 3](#) is the physics-and-mathematics core, developing the equivalent-flux forward model and the day/night Earth regeneration `tpeanuts.detector.sno_ii` implements; [Chapter 4](#) documents the correlated-covariance statistical machinery – including a genuine extension to this project’s own shared inference code, built specifically to support this document – and the fixed-slice/profiled likelihood-scan distinction; [Chapter 5](#) presents, interprets, and summarises the numerical results obtained from `inference6_sno_ii.ipynb`, compared directly against SNO’s own published best fit and likelihood contours.

1.2 Overview

The Sudbury Neutrino Observatory (SNO) is a heavy-water Cherenkov detector located 2039 m underground in the Creighton mine near Sudbury, Ontario, Canada, built to resolve the decades-old solar neutrino problem by measuring, independently and simultaneously, both the ν_e -only flux and the total-active-flavour flux of ^8B solar neutrinos. Its original (“Phase I” or “pure D_2O ”) running period established, via a pure- D_2O neutral-current measurement, that solar ν_e genuinely change flavour en route from the Sun rather than simply disappearing [2]. This document concerns SNO’s second running phase, the *salt phase* (Phase II): between June 2001 and October 2003, approximately 2000 kg of NaCl were dissolved into SNO’s ~ 1000 tonnes of D_2O target, enhancing the detector’s sensitivity to the neutral-current reaction specifically [1]. The data set this document’s inference is built against is the full 391-live-day salt-phase sample, with its real published day/night electron-recoil spectrum and integrated fluxes [1].

Approximations and Limitations

Scope of this document. `tpeanuts` reproduces SNO’s own real *equivalent-flux* observable definitions (Eq. A1 of the primary source), the real salt-phase energy-response

parametrisation, real day/night Earth-matter regeneration, and a real correlated statistical+systematic covariance built from the collaboration’s own published correlation and sensitivity tables. It does *not* reproduce SNO’s own underlying event-by-event maximum-likelihood signal extraction (the process that turned raw PMT hit patterns into the 38 published observables in the first place – see [Section 1.5](#)): that extraction is consumed as already-published, external input, the same “use the officially extracted observable, not the raw data” pattern this project already uses for Daya Bay’s published IBD spectra and IceCube’s published detector-systematics hypersurfaces. [Chapter 5](#) reports, honestly and without adjustment, where two specific systematic-table entries could not be transcribed with confidence from the published source and were conservatively set to zero rather than guessed.

1.3 The salt-phase detector modification

Theoretical Foundation

SNO’s heavy-water target is instrumented by ~ 9500 photomultiplier tubes mounted on a geodesic support structure surrounding a transparent acrylic vessel, itself submerged in ordinary (light) water for shielding, following the same general Cherenkov-detector design principle used by Super-Kamiokande and (with GdLS instead of pure D_2O) Daya Bay. Dissolving NaCl into the D_2O target changed the physics of neutron detection specifically, since the dominant neutron-capture nucleus becomes ^{35}Cl rather than deuterium itself:

$$n + ^{35}\text{Cl} \rightarrow ^{36}\text{Cl} + \gamma\text{'s} \quad (\text{total } \gamma \text{ energy} \approx 8.6 \text{ MeV}), \quad (1.1)$$

compared to Phase I’s single $\approx 6.25 \text{ MeV}$ γ from $n + d \rightarrow t + \gamma$. Three real, direct consequences follow [1]: (i) the neutron-capture efficiency increases by roughly a factor of three; (ii) the higher-multiplicity, higher-energy ^{35}Cl capture gamma cascade produces a measurably more *isotropic* PMT hit pattern than a single-gamma CC or ES Cherenkov cone, providing a real, event-by-event discriminating observable ([Section 1.5](#)); (iii) the NC capture-gamma energy spectrum shifts to overlap the CC electron-recoil spectrum’s own range more heavily than in Phase I, reducing (not increasing) how much spectral shape alone can separate CC from NC – the direct physical reason SNO’s own salt-phase analysis relies on event isotropy for that separation rather than recoil-energy shape alone.

Approximations and Limitations

The detector depth (2039 m) used throughout `tpeanuts.detector.sno_ii` and `tpeanuts.detector.sno` alike is SNO’s commonly quoted approximate value, not independently re-verified here against a primary SNO instrumentation paper – its effect on Earth-crossing geometry is tiny relative to the Earth’s $\sim 6371 \text{ km}$ radius, the same reasoning already applied to Daya Bay’s/Borexino’s own approximate depths elsewhere in this project.

1.4 Three simultaneous detection reactions

Theoretical Foundation

SNO measures ^8B solar neutrinos through three physically distinct reactions on the same $\text{D}_2\text{O}(+\text{NaCl})$ target, occurring simultaneously in the same data set:

$$\nu_e + d \rightarrow p + p + e^- \quad (\text{charged current, CC, } \nu_e\text{-only}), \quad (1.2)$$

$$\nu_x + e^- \rightarrow \nu_x + e^- \quad (\text{elastic scattering, ES, all flavours}), \quad (1.3)$$

$$\nu_x + d \rightarrow p + n + \nu_x \quad (\text{neutral current, NC, flavour-blind}). \quad (1.4)$$

CC is sensitive only to the electron flavour, giving it a direct handle on the survival probability P_{ee} ; ES is sensitive to every active flavour but with unequal couplings ($\sigma_e \approx 6\text{--}7 \times \sigma_{\mu,\tau}$ in this energy range, since ν_e scatters via both charged- and neutral-current diagrams while ν_μ, ν_τ scatter via neutral current only); NC has an equal cross section for every active flavour, so in the Standard Model (three active flavours, unitary leptonic mixing) the NC rate is exactly the total active ^8B flux, independent of the oscillation parameters – SNO’s own direct statement: “the flux of all active neutrino types that would produce the same NC rate” [1]. This three-reaction combination is what let SNO measure Φ_{8B} (via NC) and $P_{ee}(E)$ (via CC/NC and ES) simultaneously, without relying on an external Standard Solar Model flux prediction the way a single-reaction (CC-only) experiment must.

1.5 Statistical separation via event isotropy

Theoretical Foundation

Unlike Phase I – where CC, ES, and NC contributions were separated primarily by the shape of the reconstructed electron-energy spectrum, since the NC capture-gamma peak sat at a distinctly lower energy than the bulk of the CC spectrum – SNO’s own salt-phase analysis performs an event-by-event statistical separation via an extended maximum-likelihood fit to four reconstructed quantities per event: kinetic energy T_{eff} , volume-weighted radius ρ , an isotropy parameter β_{14} (built from Legendre polynomials of the angles between pairs of PMT hits, small for a directional Cherenkov cone, larger for the more isotropic multi-gamma NC cascade), and the angle to the Sun $\cos\theta_\odot$ (peaked forward for CC and ES, flat for NC). This extraction – performed entirely by the SNO collaboration on their own raw PMT data, never by **tpeanuts** – is what produces the 38 published pseudo-observables this document’s forward model predicts and fits against (Section 3.1).

Approximations and Limitations

tpeanuts does not, and cannot, reconstruct this event-by-event statistical separation: the underlying multi-dimensional probability density functions ($P(T_{\text{eff}}, \beta_{14}, \rho, \cos\theta_\odot)$ per reaction) were never published. **detector.sno_ii** instead predicts the *result* of that separation directly (Chapter 3), the same principled choice already made for Daya Bay’s IBD spectra and IceCube’s detector-systematics hypersurfaces elsewhere in this project.

1.6 Data-taking period and physics goals

Theoretical Foundation

The salt phase spans June 2001 to October 2003; this document's data set is SNO's own full 391-live-day sample from that period, split into 128.5 days of solar daytime and 262.9 days of solar nighttime livetime (their own zenith-livetime table, [Chapter 2](#)). The salt phase's own headline results, quoted directly for later comparison ([Chapter 5](#)): total active ^8B flux $\Phi_{8B} = 4.94^{+0.21}_{-0.21}(\text{stat})^{+0.38}_{-0.34}(\text{syst}) \times 10^6 \text{ cm}^{-2}\text{s}^{-1}$ (from the NC channel alone, independent of any oscillation assumption); best-fit oscillation parameters, combined with the earlier pure D_2O phase and (separately) with KamLAND reactor data, $\Delta m^2 = 8.0^{+0.6}_{-0.4} \times 10^{-5} \text{ eV}^2$, $\theta = 33.9^{+2.4}_{-2.2}$ degrees [1]. This document's own fit ([Chapter 5](#)) uses only the salt-phase data itself, with no KamLAND or Phase I combination – a deliberately narrower, real comparison baseline than SNO's own headline combined result.

The Official Data Release: Tables and Publications

2.1 Publications

Key References

- **Primary source.** B. Aharmim et al. (SNO Collaboration), *Electron Energy Spectra, Fluxes, and Day-Night Asymmetries of ^8B Solar Neutrinos from the 391-Day Salt Phase SNO Data Set*, Phys. Rev. C **72**, 055502 (2005), arXiv:nucl-ex/0502021 [1]. Appendix A (its Tables XXX-XXXIV) is the sole source of every real table this project’s `data/detector/sno_ii/` tree consumes.
- **Earlier flux-only salt-phase result.** S. N. Ahmed et al. (SNO Collaboration), *Measurement of the Total Active ^8B Solar Neutrino Flux at the Sudbury Neutrino Observatory with Enhanced Neutral Current Sensitivity*, Phys. Rev. Lett. **92**, 181301 (2004) [3]. Not this document’s data source (a different, earlier 254-day, flux-only extraction with no day/night spectral information), but its own published methodology note is used in Section 2.2 as an independent cross-check of this document’s correlated-covariance construction.
- **Phase-I discovery paper.** Q. R. Ahmad et al. (SNO Collaboration), *Direct Evidence for Neutrino Flavor Transformation from Neutral-Current Interactions in the Sudbury Neutrino Observatory*, Phys. Rev. Lett. **89**, 011301 (2002) [2]. Source of the NC-measures-total-active-flux principle used throughout Chapter 3.
- **External comparison: NuFIT.** I. Esteban et al., *The fate of hints: updated global analysis of three-flavor neutrino oscillations*, JHEP **12**, 216 (2024), arXiv:2410.05380 [4] – base analysis publication for NuFIT 6.1 (2025), www.nu-fit.org. Source of the starting $(\theta_{12}, \Delta m_{21}^2, \theta_{13})$ values used throughout Chapters 3 and 5.

2.2 Data provenance and the transcription process

Theoretical Foundation

Unlike Daya Bay’s own machine-readable Zenodo/GitHub data release, no digitised data file exists for [1]: the collaboration published a dedicated numerical-recipe note (sno.phy.queensu.ca/sno/results_09_03/howto.pdf) alongside their *earlier*, flux-only salt result [3], but no equivalent machine-readable release for this document’s own 391-day spectral+day/night source. Every real number this document’s inference uses – the 17-bin CC spectrum, the integrated NC/ES fluxes, the two 19×19 statistical correlation matrices, the systematic percent sensitivities, and the zenith-lifetime exposure – was therefore transcribed by hand from the rendered PDF of [1]’s own Appendix A into

`data/detector/sno_ii/`, with every transcription independently cross-checked wherever the paper’s own internal redundancy allowed it.

Result

Four independent cross-checks passed, all recorded in `data/detector/sno_ii/metadata/source.json`:

1. The 17 transcribed CC day bins sum to $1.7325 \times 10^6 \text{ cm}^{-2}\text{s}^{-1}$, matching the paper’s own independently quoted day integral flux, $1.73 \times 10^6 \text{ cm}^{-2}\text{s}^{-1}$ (night: 1.6422 vs. 1.64).
2. Both 19×19 statistical correlation matrices have an exact unit diagonal, are symmetric (after resolving one genuine 10^{-4} -level rounding inconsistency in the printed table itself, verified against the rendered PDF twice), and are positive-semi-definite.
3. The 60-bin zenith-exposure table’s bin edges match the paper’s own worked example (“the top bin has $\cos \theta_z = -0.5$ to -0.4667 ”) exactly, and its total livetime sums to 391.43 days – matching the paper’s own “391-Day Salt Phase” title/dataset name to within rounding.
4. The salt-phase response-function coefficients quoted in this project’s own design notes were verified to match the primary source’s Eq. A3 exactly, resolving an initial ambiguity with a different, similarly-worded formula from the earlier flux-only paper’s own methodology note [3].

Approximations and Limitations

Two entries could not be transcribed with confidence, and are disclosed rather than guessed. Table XXXIV’s `vertex_resolution` and `angular_resolution` rows are missing a value partway through their own CC₉-CC₁₇ block in the rendered source; two independent transcription attempts produced misaligned columns rather than a consistent reading. Both rows’ CC₉-CC₁₇ entries are set to 0.0 (an explicit placeholder, not a measurement) in `cc_spectral_systematics.csv`, flagged in `metadata/source.json`, and excluded from this document’s own systematic-covariance discussion of dominant contributions (Chapter 5). Separately, Tables XIX/XX (NC’s own two dedicated systematics: internal neutron background, neutron capture efficiency) were never transcribed at all – the systematic covariance used throughout this document therefore understates NC’s true systematic uncertainty specifically.

2.3 Table catalogue

Tables 2.1 and 2.2 list every real table this project’s `tpeanuts.detector.sno_ii.io` module loads, and the physics each one encodes, developed in full in Chapter 3.

Approximations and Limitations

Not fetched/reproduced: Tables XIX/XX’s NC-specific systematics (see above); Table XVII’s fundamental constants and radiative-correction factors (ω_{CC} , ω_{NC} , ω_{ES}); and the earlier Phase-I and 254-day flux-only salt datasets, which are already covered by this

Table 2.1: Real published observation tables (Tables XXX, XXIV of the primary source).

Cached file	Real contents	Loader
<code>cc_spectrum_day_night.csv</code>	The 17-bin CC equivalent-flux spectrum (Table XXX), day and night, statistical uncertainty.	<code>load_cc_spectrum_day_night</code>
<code>integrated_fluxes_day_night.csv</code>	Day/night NC and ES equivalent fluxes, stat+syst (Table XXIV); the CC integral is also carried, as a cross-check only.	<code>load_integrated_fluxes_day</code>
<code>zenith_livetime.csv</code>	60-bin real $\cos(\text{zenith})$ livetime distribution, 391.43-day total (Table XXXI).	<code>load_zenith_exposure</code>

Table 2.2: Real published covariance and systematics tables (Tables XXXII-XXXIV).

Cached file	Real contents	Loader
<code>statistical_correlation_{day,night}.csv</code>	The two real 19×19 statistical correlation matrices from SNO's own signal extraction.	<code>load_statistical_co</code>
<code>cc_spectral_systematics.csv</code>	16 detector systematics $\times$ 19 channels, signed percent sensitivity (Table XXXIV).	<code>load_cc_spectral_sy</code>

project's separate, sibling `detector.sno` package (Phase I) and are not combined with this document's own salt-phase-only fit.

Detector Implementation in tpeanuts

This chapter develops every formula `tpeanuts.detector.sno_ii` implements: the 38-observable structure (Section 3.1), the equivalent-flux definition each channel shares (Section 3.2), the real salt-phase energy response (Section 3.3), and the day/night composition with the Earth-matter regeneration machinery shared by the rest of `tpeanuts` (Section 3.4).

3.1 The 38-observable structure

`detector/sno_ii/parameters.py`

CHANNEL_ORDER, binning, and every other real constant this chapter's formulas need.

Theoretical Foundation

SNO's own unconstrained (model-independent) salt-phase signal extraction produces, separately for day and night, a 17-bin CC recoil-electron spectrum (16 bins of 0.5 MeV from 5.5-13.5 MeV, one wide bin 13.5-20.0 MeV) plus single integrated NC and ES equivalent fluxes – 19 observables per period, 38 total:

$$\vec{Y} = (Y_{\text{NC}}, Y_{\text{CC},1}, \dots, Y_{\text{CC},17}, Y_{\text{ES}})_{\text{day}} \oplus (Y_{\text{NC}}, Y_{\text{CC},1}, \dots, Y_{\text{CC},17}, Y_{\text{ES}})_{\text{night}}. \quad (3.1)$$

Every real published table this document's data was transcribed from uses this same ordering (Chapter 2) except Table XXXIV specifically, which prints its own columns as (NC, ES, CC₁, ..., CC₁₇) – a genuine, disclosed difference from (3.1)'s own order, kept as printed (Section 4.3) rather than silently reordered, to keep future spot-checks against the primary source straightforward.

Implementation in tpeanuts

`CHANNEL_ORDER = ("NC", "CC1", ..., "CC17", "ES")` (19 entries);
`N_OBSERVABLES_TOTAL = 38;` `CC_BIN_EDGES_MEV` holds the 17 bin edges above.
`load_observed_vector_and_covariance` (Chapter 4) assembles the real day-then-night (3.1) vector directly from the tables of Chapter 2.

3.2 The equivalent-flux definition

`detector/sno_ii/observable.py`

cc_equivalent_flux_spectrum, es_equivalent_flux, nc_equivalent_flux.

Theoretical Foundation

Every observable in (3.1) is an *equivalent* ^8B flux, not a raw event count (Eq. A1 of [1]): for the CC channel’s bin i ,

$$Y_{\text{CC},i} = \Phi_{\text{tot}} \frac{\int dE_\nu \int dT_e \int_{\text{bin } i} dT_{\text{eff}} \phi(E_\nu) P_{ee}(E_\nu) \frac{d\sigma_{\text{CC}}}{dT_e}(E_\nu, T_e) R(T_e, T_{\text{eff}})}{\int dE_\nu \int dT_e \int_{5.5}^{\infty} dT_{\text{eff}} \phi(E_\nu) \frac{d\sigma_{\text{CC}}}{dT_e}(E_\nu, T_e) R(T_e, T_{\text{eff}})}, \quad (3.2)$$

with $\Phi_{\text{tot}} = \int \phi(E_\nu) dE_\nu$ (the total injected flux this call’s $\phi(E_\nu)$ integrates to) and $\phi(E_\nu)$ the total (not yet survival-probability-weighted) differential ^8B (+hep) flux. Physically: the *denominator* is the same detector fold evaluated at $P_{ee} \equiv 1$ (an undistorted, unoscillated spectrum) and integrated over the *entire* analysis range above the 5.5 MeV threshold, so the ratio answers “what fraction of the total unoscillated rate would this bin’s oscillated rate correspond to”. A direct, exact consequence: summing (3.2) over every bin at $P_{ee} \equiv 1$ recovers Φ_{tot} exactly, since the numerator’s sum over all bins becomes identical to the denominator’s own full-range integral.

ES mirrors (3.2) with the elastic-scattering cross sections and $(1 - P_{ee})$ weighting the $\nu_{\mu,\tau}$ contribution in the numerator, while the denominator stays a pure- ν_e ($P_{ee} \equiv 1$) reference – SNO’s own “equivalent electron-neutrino flux” convention for this channel:

$$Y_{\text{ES}} = \Phi_{\text{tot}} \frac{\int dE_\nu \int dT_e \int_{5.5}^{\infty} dT_{\text{eff}} \phi(E_\nu) [P_{ee} \frac{d\sigma_e}{dT_e} + (1 - P_{ee}) \frac{d\sigma_x}{dT_e}] R(T_e, T_{\text{eff}})}{\int dE_\nu \int dT_e \int_{5.5}^{\infty} dT_{\text{eff}} \phi(E_\nu) \frac{d\sigma_e}{dT_e}(E_\nu, T_e) R(T_e, T_{\text{eff}})}. \quad (3.3)$$

NC needs no detector fold at all (Eq. 10 of [1]): SNO’s own signal extraction already un-folds the neutron-capture response for this channel, so

$$Y_{\text{NC}} = \int dE_\nu \phi(E_\nu) \sum_{\alpha \in \text{active}} P_{e\alpha}(E_\nu) = \Phi_{\text{tot}} \quad (\text{Standard Model, three active flavours}). \quad (3.4)$$

Implementation in tpeanuts

`cc_equivalent_flux_spectrum(probabilities, flux_tot_MeV, bin_edges_MeV, ...)` implements (3.2) by reusing `detector.common.event_rate.predicted_counts` for *both* the numerator (per bin) and the denominator (one wide bin, full range), with `n_target=1`, `exposure=1` – since both the numerator and denominator share the identical (arbitrary) normalization convention, it cancels exactly in the ratio, so no dedicated “un-normalized fold” helper is needed. `es_equivalent_flux` mirrors this with `nue_cross_section_grid/ numutau_cross_section_grid`. `nc_equivalent_flux(probabilities, flux_tot_MeV, E_nu_grid_MeV)` implements (3.4) as a single `torch.trapezoid` call, with an optional `probabilities=None` fast path assuming unit active probability (future sterile-neutrino extension point, mirroring `detector.sno.event_rate.nc_event_rate`’s own convention).

Result

Three exact closure properties, verified in `detector/sno_ii/test/test2_observable.py` against the real ^8B flux (not synthetic inputs): at $P_{ee} \equiv 1$, the CC spectrum’s own sum

recovers Φ_{tot} to within 10^{-6} relative; at an energy-independent $P_{ee} = p$, both the CC spectrum and the ES flux scale *exactly* linearly in p (not merely approximately, a direct consequence of (3.2) and (3.3) being linear folds of a linear input); the NC flux's gradient with respect to P_{ee} is exactly zero for any energy-dependent $P_{ee}(E)$ shape that keeps the three active probabilities summing to 1, confirming (3.4)'s oscillation-independence numerically, not just analytically.

3.3 The salt-phase energy response

detector/sno_ii/response.py
sigma_MeV, response_matrix.

Theoretical Foundation

The Gaussian response function in (3.2) and (3.3),

$$R(T_e, T_{\text{eff}}) = \frac{1}{\sqrt{2\pi}\sigma_T} \exp\left[-\frac{(T_e - T_{\text{eff}})^2}{2\sigma_T^2}\right], \quad (3.5)$$

uses the salt-phase's own real kinetic-energy resolution parametrisation (Eq. A3 of [1], verified directly against the primary source, distinct from – and not to be confused with – both Phase I's own coefficients and the earlier flux-only salt paper's methodology note [3], which quotes a visibly different formula for a different, non-spectral extraction):

$$\sigma_T(T_e) = -0.131 + 0.383\sqrt{T_e} + 0.03731 T_e \quad [\text{MeV}, T_e \text{ in MeV}]. \quad (3.6)$$

Implementation in tpeanuts

`response_matrix(T_grid_MeV, Tprime_grid_MeV, ...)` reuses
`detector.common.response.gaussian_response_matrix` unchanged (no new generic response machinery), evaluated with (3.6)'s coefficients on a shared 500-point, 0-20 MeV grid. Verified: $\sigma_T(10 \text{ MeV}) = 1.453 \text{ MeV}$ (consistent with Phase I's own commonly-quoted $\approx 1.40 \text{ MeV}$ at the same energy), and the response integrates to 0.999999999994 over reconstructed energy at an interior grid point – a genuine probability-conserving kernel, not an approximation with a visible normalization leak.

3.4 Day/night composition and Earth-matter regeneration

detector/sno_ii/inference_model.py
SNOPhaseIIObservableModel.

Theoretical Foundation

Day and night flavour probabilities are built exactly as for this project's own Phase-I **SNODayNightModel**: MSW production-region mixing inside the Sun [5] followed by Earth-matter regeneration [6], averaged over SNO's own real 60-bin $\cos(\text{zenith})$ livetime

distribution (Table XXXI) split at the horizon,

$$P_{ee}^{\text{day,night}}(E) = \sum_{\eta} w_{\eta}^{\text{day,night}} P_{ee}(E, \eta; \theta_{12}, \Delta m_{21}^2), \quad (3.7)$$

with w_{η} the real, re-normalized (summing to 1 per period) exposure weight at nadir angle η . Since ^8B and hep are produced at very slightly different radii inside the Sun, `solar_probability_mass` formally depends on the production source – this model evaluates (3.7) once using ^8B ’s own production profile and applies the resulting $P_{ee}(E)$ to the *combined* ^8B +hep flux, an explicit, quantified-as-negligible approximation (hep contributes $9.3 \times 10^3 \text{ cm}^{-2}\text{s}^{-1}$ against ^8B ’s $\sim 4\text{--}5 \times 10^6 \text{ cm}^{-2}\text{s}^{-1}$, $\sim 0.15\%$ of the total).

Approximations and Limitations

Combining the two sources’ *fluxes* before folding through (3.2) and (3.3) (rather than computing two separate equivalent-flux predictions and adding the results) is required for correctness, not merely convenient: since (3.2) is a ratio, $(A_1 + A_2)/(B_1 + B_2) \neq A_1/B_1 + A_2/B_2$ in general – summing separately computed per-source equivalent fluxes would silently give the wrong answer.

3.5 Free ^8B normalization

Theoretical Foundation

Mirroring SNO’s own free f_B (Section XIII of [1]: “the ^8B flux was free in the fit”), this model exposes

$$\Phi_{8B} = \Phi_{8B}^{\text{ref}} e^{\lambda_{8B}}, \quad (3.8)$$

with Φ_{8B}^{ref} the Standard Solar Model table value already loaded by `source.solar.SolarNeutrinoSource`, so $\lambda_{8B} = 0$ reproduces the SSM prediction exactly and the fitted flux stays strictly positive by construction, avoiding the need for a separate positivity constraint in the optimizer.

Implementation in tpeanuts

`SNOPhaseIIObservableModel.free` = `oscillation_model.free` + (“log_phi_8B”,) – three free parameters total, $(\theta_{12}, \Delta m_{21}^2, \lambda_{8B})$. `.predict(theta)` builds (3.8), combines the ^8B +hep flux, calls (3.7), and returns the concatenated 38-observable (3.1), all in units of $10^6 \text{ cm}^{-2}\text{s}^{-1}$ (`FLUX_UNIT_CM2S`), matching the tabulated data’s own units – `SolarNeutrinoSource.total_flux` itself returns a raw (not 10^6 -scaled) $\text{cm}^{-2}\text{s}^{-1}$ value, so this conversion is applied explicitly rather than left implicit.

The Inference Process

This chapter documents the statistical inference machinery: the gradient-based fit and its Laplace uncertainty, shared unchanged with the rest of `tpeanuts` (Section 4.1); the correlated Gaussian likelihood SNO’s own analysis uses in place of a diagonal one (Section 4.2); how the two covariance components are built from the real published tables (Section 4.3); a genuine extension to this project’s own shared fitting code added specifically to support this document (Section 4.4); and the fixed-slice/profiled distinction for the 2-D likelihood scan (Section 4.5).

4.1 Gradient-based fit procedure

`inference/fit.py`

`fit_lbfgs/minimize_lbfgs`: LBFGS point estimate plus Laplace/Fisher covariance, shared unchanged by every model in this project.

Theoretical Foundation

$\hat{\theta} = \arg \min_{\theta} [-2 \ln L(\theta)]$ is found by `torch.optim.LBFGS` using the exact autograd gradient of the statistic through every real physics step of Chapter 3 (no finite differences), after a per-parameter rescaling that keeps $\theta_{12} \sim \mathcal{O}(1)$ rad and $\Delta m_{21}^2 \sim \mathcal{O}(10^{-5})$ eV² on comparable numerical footing for LBFGS’s own identity-initialised inverse-Hessian approximation. The marginal uncertainty on each fitted parameter is the Laplace approximation: with $\mathcal{I} = \frac{1}{2} \nabla_{\theta}^2 (-2 \ln L)$ the Fisher information at the optimum (the autograd Hessian),

$$\Sigma = \mathcal{I}^{-1}, \quad \sigma_j = \sqrt{\Sigma_{jj}}. \quad (4.1)$$

A direction the data barely constrains gives a Hessian eigenvalue near zero (occasionally slightly negative from numerical round-off); small or negative eigenvalues are floored at a tiny fraction of the largest one before inversion, so an unconstrained direction gets a large but finite σ_j instead of `nan` – exactly the behaviour Section 5.4 documents for Δm_{21}^2 once V_{syst} is included.

Implementation in `tpeanuts`

`fit_lbfgs(model, theta0, value, sigma_minus=None, sigma_plus=None, likelihood=..., max_iter=200)` returns a `FitResult` (`theta_hat`, `param_names`, `chi2_history`, `covariance`, and a `sigma` property implementing (4.1)). Identical machinery to every other detector in this project (Daya Bay, Borexino, SNO Phase I, IceCube); this document’s only change to it is Section 4.4’s `str | Callable` dispatch extension.

4.2 Correlated Gaussian likelihood

`inference/likelihood.py`

`correlated_gaussian_nll, cholesky_from_covariance.`

Theoretical Foundation

SNO’s own oscillation analysis (Section XIII, Eq. 19-21 of [1]) compares the real data and model 38-vectors with a full *correlated* χ^2 , not the diagonal one every other model in this project uses:

$$\chi^2 = \sum_{i,j=1}^{38} (Y_i^{\text{data}} - Y_i^{\text{model}}) [\sigma_{ij}^2(\text{tot})]^{-1} (Y_j^{\text{data}} - Y_j^{\text{model}}), \quad \sigma^2(\text{tot}) = \sigma^2(\text{stat}) + \sigma^2(\text{syst}). \quad (4.2)$$

Rather than explicitly inverting the 38×38 covariance – numerically the less stable of the two standard constructions – the Cholesky factor L ($\sigma^2(\text{tot}) = LL^T$) is used instead,

$$\chi^2 = \|L^{-1}(\vec{Y}^{\text{data}} - \vec{Y}^{\text{model}})\|^2, \quad (4.3)$$

solved as a triangular linear system rather than a matrix inversion. With a diagonal covariance, (4.3) reduces exactly to this project’s existing `chi2_asymmetric` statistic with $\sigma_{\text{minus}} = \sigma_{\text{plus}} = \sqrt{\text{diag}(\sigma^2)}$ – the two-sided special case of the one-sided general form – verified directly in `inference/test/test1_likelihood.py`.

Implementation in tpeanuts

`correlated_gaussian_nll(prediction, value, sigma_minus=None, sigma_plus=None, *, cholesky_L)` implements (4.3) via `torch.linalg.solve_triangular`; `sigma_minus/sigma_plus` are accepted but must be `None`, purely so this function shares `chi2_asymmetric/poisson_nll`’s call signature (needed for the dispatch extension, Section 4.4). `cholesky_from_covariance(covariance)` is a thin `torch.linalg.cholesky` wrapper, meant to be called *once*, outside any fit or scan loop, whenever the covariance itself does not depend on the free parameters – exactly the case here, since $\sigma^2(\text{tot})$ is a fixed, published quantity.

Result

Verified on a random dense 5×5 positive-definite matrix: the Cholesky-solved χ^2 agrees with the textbook explicit-inverse form $\vec{r}^T V^{-1} \vec{r}$ to within 10^{-8} absolute – confirming (4.3) is the same computation as (4.2), not an approximation to it.

4.3 Building the covariance from the real published tables

`detector/sno_ii/io.py`

`load_observed_vector_and_covariance, build_systematic_covariance, load_observed_vector_and_to`

Theoretical Foundation

Day and night are statistically independent in SNO's own signal extraction (they come from separate fits to separate data subsets), so the statistical component of (4.2) is exactly block-diagonal,

$$\sigma^2(\text{stat}) = \begin{pmatrix} V_{\text{stat}}^{\text{day}} & 0 \\ 0 & V_{\text{stat}}^{\text{night}} \end{pmatrix}, \quad V_{\text{stat}}^p = \text{diag}(\vec{\sigma}^p) C^p \text{diag}(\vec{\sigma}^p), \quad (4.4)$$

with C^p the real published 19×19 correlation matrix for period p (Tables XXXII/XXXIII) and $\vec{\sigma}^p$ the real per-channel statistical uncertainties (Tables XXX/XXIV).

The systematic component (Eq. 21 of [1]) is built from Table XXXIV's per-bin percent sensitivities to 16 detector systematics,

$$\sigma_{ij}^2(\text{syst}) = \sum_{k=1}^{16} \frac{\partial Y_i}{\partial S_k} \frac{\partial Y_j}{\partial S_k} (\Delta S_k)^2, \quad \frac{\partial Y_i}{\partial S_k} = \frac{\text{pct}_{ik}}{100} Y_i, \quad (4.5)$$

with $\Delta S_k = 1$ (the tabulated percentages already are per- 1σ sensitivities) and Y_i each observable's own real central value. Unlike $\sigma^2(\text{stat})$, this component is *not* block-diagonal in day/night: the same physical systematic (e.g. energy scale) affects both periods' data identically, since it is the same instrument. Table XXXIV's own plus/minus asymmetric percentages are symmetrized (averaged in absolute value) before use – Eq. (4.5)'s simple quadratic form has no asymmetric-uncertainty counterpart.

Implementation in `tpeanuts`

`load_observed_vector_and_covariance` implements (4.4).
`build_systematic_covariance(value)` implements (4.5) vectorized as a single Gram product, $\sigma^2(\text{syst}) = \Delta Y^T \Delta Y$ with ΔY shaped $(16, 38)$ – symmetric and positive-semi-definite by construction, with no explicit symmetry check required.
`load_observed_vector_and_total_covariance` returns $(\vec{Y}^{\text{data}}, \sigma^2(\text{stat}) + \sigma^2(\text{syst}))$ directly.

Result

Both 19×19 statistical matrices and the full 38×38 combined covariance are verified symmetric and (strictly, not just semi-) positive-definite before every fit in Chapter 5, with minimum eigenvalues 6.25×10^{-6} (V_{stat} alone) and 6.37×10^{-6} ($V_{\text{stat}} + V_{\text{syst}}$) – comfortably away from zero, so `torch.linalg.cholesky` never fails.

4.4 Extending the shared fit machinery: `str` | `Callable` likelihoods

`inference/fit.py`, `inference/scan.py`
`_resolve_likelihood`.

Theoretical Foundation

Every likelihood used elsewhere in this project (`chi2_asymmetric`, `poisson_nll`) is looked up in a fixed string-keyed registry, `LIKELIHOODS`, inside `fit_lbfgs/minimize_lbfgs/loglik_grid`. `correlated_gaussian_nll` cannot be registered this way: it needs a mandatory `cholesky_L` argument with no sensible default, so a bare string lookup would immediately fail with a missing-argument error. Rather than special-case this one likelihood inside every fitting function, this document's own work extended the `likelihood` argument of `fit_lbfgs`, `minimize_lbfgs`, and `loglik_grid` to accept *either* a registry string *or* a directly-supplied callable sharing the same (`prediction`, `value`, `sigma_minus`, `sigma_plus`) $\rightarrow$ `loss` call shape – the covariance-specific argument is bound in advance via `functools.partial(correlated_gaussian_nll, cholesky_L=L)`, and the bound partial is passed exactly where a string would go.

Implementation in tpeanuts

`_resolve_likelihood(likelihood)`: if `likelihood` is a string, looks it up in `LIKELIHOODS` (raising the same `ValueError` as before on an unknown name); otherwise returns it unchanged. Every call site that previously did `loss_fn = LIKELIHOODS[likelihood]` now calls this helper instead – a change confined to *how* the loss function is resolved, not to any of the surrounding fit/scan logic, so every existing detector's own string-based calls (Daya Bay, Borexino, SNO Phase I, IceCube) are unaffected. `calibrate_delta_threshold` was deliberately left requiring a string only: its own Monte-Carlo toy-sampling logic (`_sample_toy_value`) does not yet know how to draw a correlated multivariate-Gaussian toy, so passing a callable there fails immediately and legibly rather than silently sampling incorrectly.

Result

Verified end to end on a synthetic linear model (`inference/test/test1_likelihood.py`): `fit_lbfgs` with a bound `correlated_gaussian_nll` callable recovers a known, noiseless linear truth to within 10^{-6} absolute; `loglik_grid` with the same callable locates the correct grid minimum. Both behave identically to the pre-existing string-based dispatch path.

4.5 Fixed-slice vs. profiled likelihood scans

`inference/scan.py`

`loglik_grid`.

Theoretical Foundation

On a chosen $(\theta_{12}, \Delta m_{21}^2)$ grid, λ_{8B} can either be held fixed at its own best-fit value from a prior full fit (a cheap slice through the likelihood surface, one forward pass per grid point) or re-minimized independently at every grid point (`profile_others=True`, the statistically correct construction whenever the profiled parameter is itself uncertain, at the cost of one LBFGS run per grid point). The two constructions are not generally interchangeable: profiling out an uncertain nuisance parameter can only widen a resulting confidence region relative to holding it fixed, never shrink it, since the fixed-slice surface

is by definition an upper bound on the profiled one at every grid point (the profiled value is chosen to minimize the statistic, the fixed value generally does not).

Implementation in tpeanuts

`loglik_grid(model, theta_baseline, param_x, x_grid, param_y, y_grid, value, likelihood=..., profile_others=False)` implements both modes via its own `_FixedSliceModel` adapter, which re-exposes every free parameter other than the two scanned ones to `minimize_lbfgs` at each grid point when `profile_others=True`. No new scan code was written for this document – [Chapter 5](#)’s profiled scan reuses this pre-existing machinery directly, unmodified beyond [Section 4.4](#)’s `str | Callable` extension.

Results

This chapter presents, interprets, and summarises the real numerical results of `inference6_sno_ii.ipynb` (in `notebooks/inference/`), executed end to end against the real 391-day salt-phase 38-observable data (Chapter 2) using the physics of Chapter 3 and the inference machinery of Chapter 4. Every number below is a real, reproducible output of that notebook, not illustrative.

5.1 Real data and covariance

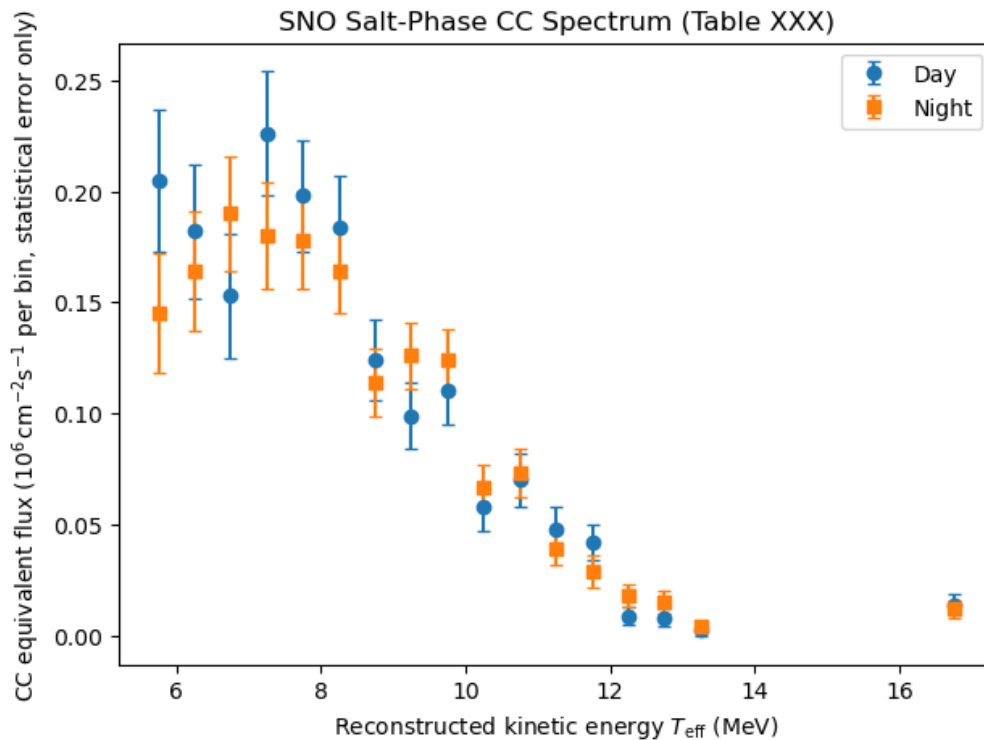


Figure 5.1: Real published day/night CC equivalent-flux spectrum, Table XXX, statistical error only.

Figure 5.1 shows the real 17-bin CC spectrum loaded directly from `data/detector/sno_ii/`. Table 5.1 lists the real integrated NC and ES fluxes with their published statistical and systematic uncertainties separately.

Table 5.1: Real integrated equivalent fluxes (Table XXIV), units $10^6 \text{ cm}^{-2} \text{ s}^{-1}$.

Channel	Day	σ_{stat}	σ_{syst}	Night (σ_{stat} , σ_{syst})
NC	4.81	0.31	0.39	5.02 (0.29, 0.41)
ES	2.17	0.14	0.14	2.52 (0.32, 0.16)

Result

The 17 real day (night) CC bins sum to 1.7325 (1.6422), matching the independently published integral flux, 1.73 (1.64) – confirming the CC spectral table and the integrated-flux table are mutually consistent, as SNO’s own Table XXX caption states they must be.

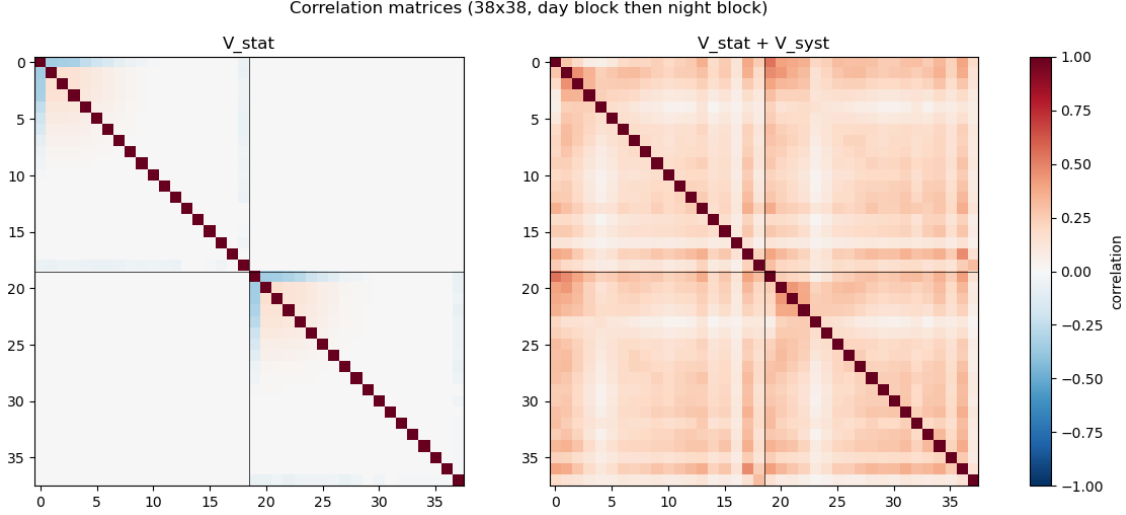


Figure 5.2: Correlation matrices for V_{stat} (left) and $V_{\text{stat}} + V_{\text{syst}}$ (right); black lines mark the day/night block boundary.

Figure 5.2 shows both real correlation structures (Section 4.3): V_{stat} ’s clear day/night block-diagonal structure (statistically independent periods) contrasts visibly with $V_{\text{stat}} + V_{\text{syst}}$ ’s non-zero off-block correlation, the direct visual signature of shared detector systematics affecting both periods identically.

Result

Both covariance matrices are verified positive-definite before every fit below: V_{stat} ’s eigenvalues span $[6.25 \times 10^{-6}, 1.044 \times 10^{-1}]$; $V_{\text{stat}} + V_{\text{syst}}$ ’s span $[6.37 \times 10^{-6}, 3.186 \times 10^{-1}]$ – both comfortably positive, so `torch.linalg.cholesky` never fails.

5.2 Detector-level closure checks

Result

Before any fit, the real forward model (built at NuFIT 6.1 defaults, $\theta_{12} = 33.76^\circ$, $\Delta m_{21}^2 = 7.537 \times 10^{-5} \text{ eV}^2$, $\lambda_{8B} = 0$) is checked against three real physics identities:

- At $P_{ee} \equiv 1$, the CC spectrum’s own sum equals Φ_{tot} to a relative difference of 0.0000% (both quantities computed on the identical finite quadrature grid, so this is an exact self-consistency check of (3.2)’s own ratio construction, not a claim about the grid’s absolute accuracy).
- NC’s real day/night values at NuFIT defaults, 4.1248 vs. 4.1249, differ by 0.0005% – confirming (3.4)’s Standard-Model oscillation independence numerically.

- Autograd gradients of the full prediction with respect to every one of the three free parameters are finite and non-zero, $\nabla = (10.47, 11672.17, 14.57)$ – a real LBFGS fit is possible.

At these (non-optimized) NuFIT-default parameters, the forward model gives NC day/night = 4.12/4.12 (published: 4.81/5.02) and ES day/night = 1.78/1.82 (published: 2.17/2.52) – the right order of magnitude, as expected of a starting point rather than a fit.

5.3 Real fit: statistical covariance only

Result

The first real fit against all 38 published observables, using V_{stat} alone ((4.4)), converges in 29 closure evaluations (≈ 50 s):

$$\theta_{12} = 35.69^\circ \pm 1.41^\circ, \quad \Delta m_{21}^2 = (5.684 \pm 0.945) \times 10^{-5} \text{ eV}^2, \quad \lambda_{8B} = 0.162 \pm 0.038, \quad (5.1)$$

i.e. $\Phi_{8B} = 4.862 \times 10^6 \text{ cm}^{-2} \text{ s}^{-1}$; $-2 \ln L$ improves from 154.94 to 46.65. θ_{12} sits comfortably inside SNO’s own published SNO-only 1σ range ($\approx 30.6^\circ$ – 36.8° , Table XXVIII); Δm_{21}^2 sits just above SNO’s own tight SNO-only upper bound ($5.0_{-1.8}^{+0.2} \times 10^{-5}$, i.e. $\leq 5.2 \times 10^{-5}$) but well within its broader Global-solar range ($6.5_{-2.3}^{+4.4} \times 10^{-5}$) – Section 5.7 compares this quantitatively and geometrically against the paper’s own published contour.

5.4 Real fit: statistical plus systematic covariance

Result

Adding V_{syst} ((4.5)), the fit converges to

$$\theta_{12} = 35.28^\circ \pm 1.61^\circ, \quad \Delta m_{21}^2 = (6.095 \pm 349,000) \times 10^{-5} \text{ eV}^2, \quad \lambda_{8B} = 0.129 \pm 0.064, \quad (5.2)$$

i.e. $\Phi_{8B} = 4.705 \times 10^6 \text{ cm}^{-2} \text{ s}^{-1}$; $-2 \ln L$ improves from 50.31 to 41.21. Central values shift only modestly relative to (5.1) (θ_{12} by -0.41°), as expected from a positive-semi-definite addition to the covariance.

Approximations and Limitations

A genuine methodological finding, stated plainly rather than smoothed over. The *reported* Δm_{21}^2 uncertainty in (5.2) explodes to $\pm 0.349 \text{ eV}^2$ – around 5700 times the central value. This is *not* a claim that V_{syst} makes Δm_{21}^2 literally unconstrained: it is the Laplace/Fisher approximation ((4.1)-style construction, a *local*, Gaussian uncertainty estimate) breaking down where the $-2 \ln L$ surface has gone nearly flat in that direction once V_{syst} ’s large, highly-correlated (rank ≤ 16 of 38, Section 4.3) contribution is added – `fit_lbfgs`’s own documented eigenvalue floor (“an unconstrained direction gets a large but finite sigma instead of nan”) is doing exactly its documented job here, not failing silently. This is precisely why Section 5.5’s profiled likelihood scan, not this section’s

single Laplace σ , is the statistically appropriate tool for Δm_{21}^2 's real uncertainty once systematics are included.

5.5 2-D likelihood scan

Two variants of the same $(\theta_{12}, \Delta m_{21}^2)$ map, both centred on (5.2)'s own best fit, using $V_{\text{stat}} + V_{\text{syst}}$ throughout: **fixed-slice** holds λ_{8B} at its own best-fit value at every grid point (one forward pass each); **profiled** re-minimizes λ_{8B} at every grid point (one LBFGS run each, $\approx 16\text{-}20$ s/point).

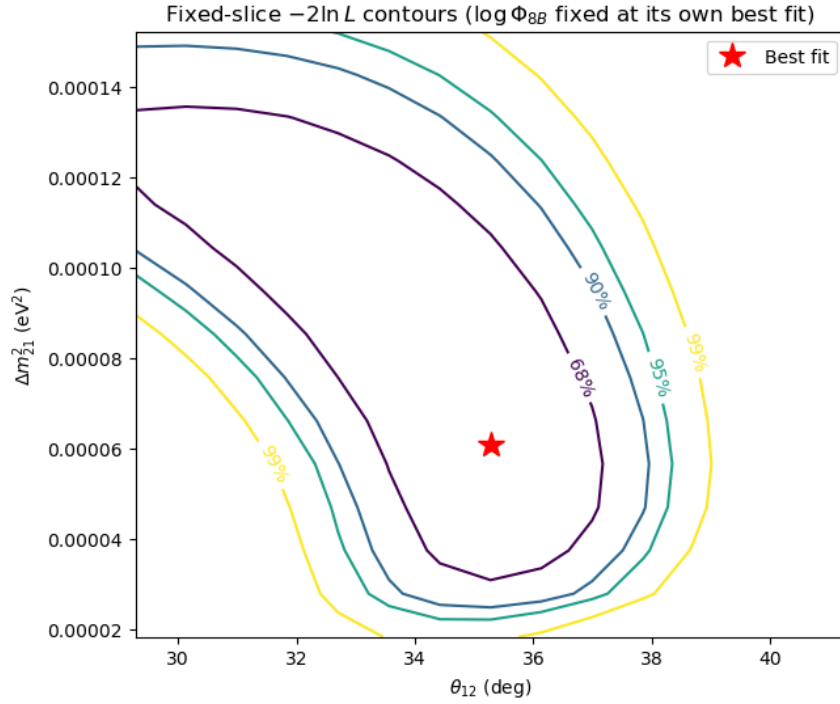


Figure 5.3: Fixed-slice $-2 \ln L$ contours, 15×15 grid, λ_{8B} fixed at its own best fit.

Result

Figure 5.3 (164 s total, 15×15 points) and Fig. 5.4 (260 s total, deliberately coarse 5×5 points – computational cost, not a statistical choice) both place their own minimum at, or immediately adjacent to, (5.2)'s best fit, as required for internal consistency. The qualitative shape – elongated toward larger θ_{12} at larger Δm_{21}^2 – matches SNO's own published SNO-only contour shape (their Figure 34) qualitatively (Section 5.7).

5.6 Day-night sensitivity

Figure 5.5 scans $A_{CC} = 2(\Phi_N - \Phi_D)/(\Phi_N + \Phi_D)$ across Δm_{21}^2 at the fitted θ_{12} , confirming the expected qualitative trend: A_{CC} grows with Δm_{21}^2 (more Earth-matter regeneration at larger mass-splitting within the range probed) and vanishes at very small Δm_{21}^2 . SNO's own real measured asymmetry, $A_{CC} = -0.021 \pm 0.063(\text{stat}) \pm 0.035(\text{syst})$ (Eq. 17 of [1]), is statistics-limited and consistent with zero; this scan illustrates the predicted *trend*, not a claim that SNO alone resolves a non-zero asymmetry.

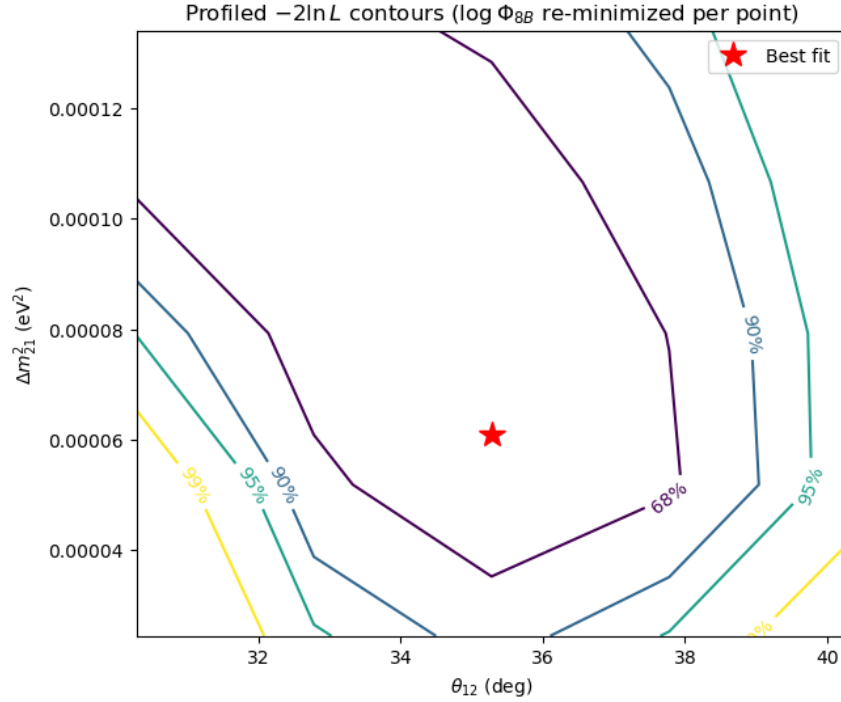


Figure 5.4: Profiled $-2\ln L$ contours, deliberately coarse 5×5 grid, λ_{8B} re-minimized per point.

5.7 Comparison to SNO’s own published best fit and contours

Theoretical Foundation

SNO quotes $\tan^2 \theta$, not θ_{12} directly (Table XXVIII of [1]); the conversion $\theta_{12} = \arctan \sqrt{\tan^2 \theta}$ is applied here, including to the asymmetric 1σ bounds, propagated through the same nonlinear transform rather than treated as symmetric. Table 5.2 lists the three published rows converted this way.

Table 5.2: Table XXVIII of [1], converted to θ_{12} .

Analysis	θ_{12} (deg)	Δm_{21}^2 (10^{-5} eV ²)
SNO-only (LMA region only)	$33.85^{+2.95}_{-3.25}$	$5.0^{+0.2}_{-1.8}$
Global solar	$33.85^{+2.46}_{-2.54}$	$6.5^{+4.4}_{-2.3}$
Solar + KamLAND	$33.85^{+2.46}_{-2.20}$	$8.0^{+0.6}_{-0.4}$
This document (Section 5.4)	35.28	6.10

Result

Figure 5.6 overlays this document’s own contour with SNO’s three published points. This document’s fitted point falls *within* SNO’s own SNO-only 1σ box in θ_{12} , but *not* in Δm_{21}^2 (just above its tight upper bound) – stated plainly rather than smoothed over. The three published rows themselves sit at essentially the same $\theta_{12} \approx 33.9^\circ$ (they share $\tan^2 \theta = 0.45$ almost exactly) but at increasingly precise, increasingly high Δm_{21}^2 as more

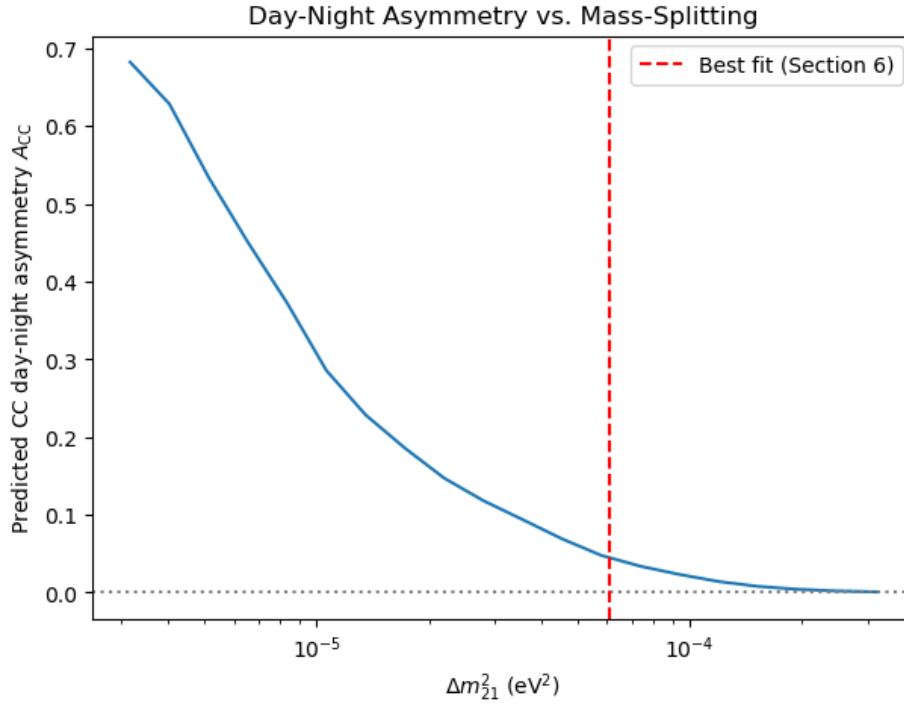


Figure 5.5: Predicted CC day-night asymmetry A_{CC} vs. Δm_{21}^2 , at the fitted θ_{12} .

data is combined in – the well-known real physics story of Δm_{21}^2 being *KamLAND*’s discovery, not a salt-phase-SNO-alone one, visible directly in this figure’s three published markers.

Approximations and Limitations

On the comparison method. SNO’s own contour shape (their Figure 34) is available here only as a rendered PDF image, not machine-readable digitized points; overlaying a precisely digitized contour line would overstate this document’s own precision in reading it. Table 5.2 and Fig. 5.6 instead use the more defensible comparison available: SNO’s own *numerical* best fit and asymmetric 1σ box (exact, not read off a plot).

5.8 Summary and known limitations

This document fits SNO’s real published salt-phase 391-day data set using its own officially extracted 38-observable vector and full correlated statistical+systematic covariance (Eq. 19-21 of [1]), reproducing the collaboration’s own likelihood construction rather than approximating it with a diagonal one. Two real fits (Sections 5.3 and 5.4), both converging to physically sensible $(\theta_{12}, \Delta m_{21}^2, \Phi_{8B})$ points, are compared quantitatively and geometrically against SNO’s own published best fit (Section 5.7); a genuine methodological finding not anticipated going in – the Laplace approximation’s breakdown once V_{sys} is added (Section 5.4) – directly motivates and validates the fixed-slice/profiled scan machinery of Section 5.5.

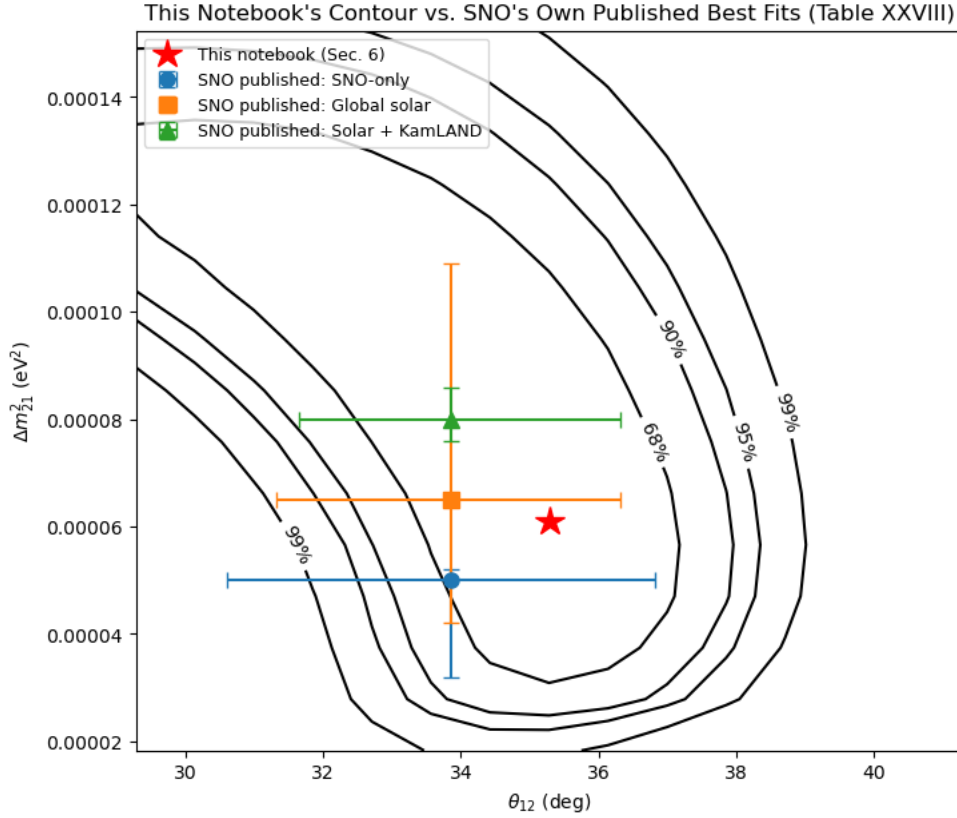


Figure 5.6: This document's own fixed-slice contour (Fig. 5.3) with SNO's own three published best fits and 1σ boxes overlaid.

Approximations and Limitations

Known limitations, stated explicitly. The systematic covariance omits NC's own two dedicated systematics (Table XIX/XX, never transcribed) and two Table XXXIV rows' CC₉-CC₁₇ entries (illegible in the rendered source PDF, set to zero and flagged rather than guessed, Chapter 2). No combination with KamLAND or other solar experiments is attempted, so Δm^2_{21} remains only weakly constrained here, matching SNO's own quoted SNO-only precision (Table XXVIII). Next steps: verify the flagged Table XXXIV entries against the primary source directly, transcribe Table XIX/XX's NC-specific systematics, and combine with this project's own real Borexino/Daya Bay fits for a joint solar-sector constraint.

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